Lab 7: Fuzzing for Input Validation Bugs

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Assigned repo: SETAPESU26/L6_L7_40_ride_share_fare (Lab 7)

Submission: github.com/menonrhea2005-rgb/PES1UG24AM222_SubscriptionBoxScheduler/tree/main/Lab7_Fuzz_Testing

The application under test is `ride_share_fare.py` (the handout's `processor.py`), with three functions: `parse_trip`, `fare_average` and `payable_fare`. The tests are in `test_ride_share_fare.py` (the handout's `test_processor.py`).

Step 1: Installing dependencies

```
py -m pip install hypothesis pytest
```

```
Requirement already satisfied: iniconfig>=2 in c:\users\arram\appdata\local\programs\python\python313\lib\site-packages
(from pytest) (2.3.1)
Requirement already satisfied: packaging>=24 in c:\users\arram\appdata\local\programs\python\python313\lib\site-packages
 (from pytest) (26.3)
Requirement already satisfied: pluggy<2,>=1.6 in c:\users\arram\appdata\local\programs\python\python313\lib\site-package
s (from pytest) (1.7.0)
Requirement already satisfied: pygments>=2.15 in c:\users\arram\appdata\local\programs\python\python313\lib\site-package
s (from pytest) (2.21.0)
Downloading hypothesis-6.170.15-cp313-cp313-win_amd64.whl (691 kB)
   ---------------------------------------- 691.4/691.4 kB 7.9 MB/s eta 0:00:00
Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl (29 kB)
Installing collected packages: sortedcontainers, hypothesis
  WARNING: The script hypothesis.exe is installed in 'C:\Users\arram\AppData\Local\Programs\Python\Python313\Scripts' wh
ich is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed hypothesis-6.170.15 sortedcontainers-2.4.0

[notice] A new release of pip is available: 24.3.1 -> 26.3
[notice] To update, run: C:\Users\arram\AppData\Local\Programs\Python\Python313\python.exe -m pip install --upgrade pip
PS C:\Users\arram\Downloads\PES1UG24AM222_SubscriptionBoxScheduler\Lab7_Fuzz_Testing>
```

Hypothesis 6.170.15 installed; pytest was already installed.

Step 2: Running the buggy application

```
py ride_share_fare.py
```

```
==== Running fare_average ====
Enter fares as CSV (e.g. 120,95,210): Average Fare: 141

==== Running payable_fare ====
Enter fare|promo (e.g. 250|30): Payable Fare: 265.0
PS C:\Users\arram\Downloads\PES1UG24AM222_SubscriptionBoxScheduler\Lab7_Fuzz_Testing>
```

```
PS C:\Users\arram\Downloads\PES1UG24AM222_SubscriptionBoxScheduler\Lab7_Fuzz_Testing> "sedan" | py ride_share_fare.py
==== Running parse_trip ====
Enter cab:distance in km (e.g. sedan:12): Traceback (most recent call last):
  File "C:\Users\arram\Downloads\PES1UG24AM222_SubscriptionBoxScheduler\Lab7_Fuzz_Testing\ride_share_fare.py", line 31,
in <module>
    main()
    ~~~~^^
  File "C:\Users\arram\Downloads\PES1UG24AM222_SubscriptionBoxScheduler\Lab7_Fuzz_Testing\ride_share_fare.py", line 17,
in main
    result = parse_trip(user_input)
  File "C:\Users\arram\Downloads\PES1UG24AM222_SubscriptionBoxScheduler\Lab7_Fuzz_Testing\ride_share_fare.py", line 2, i
n parse_trip
    cab, km = text.split(':')
    ^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
PS C:\Users\arram\Downloads\PES1UG24AM222_SubscriptionBoxScheduler\Lab7_Fuzz_Testing>
```

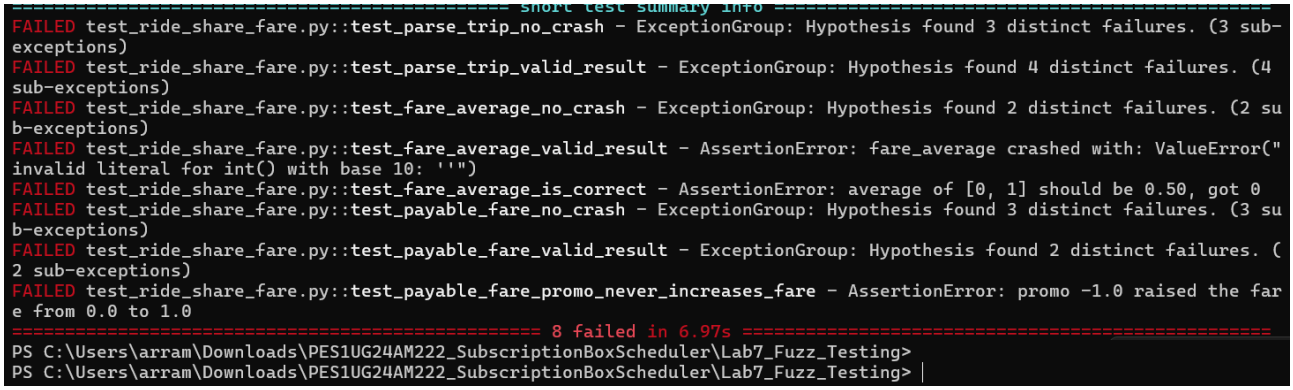
With valid input (`sedan:12`, `120,95,210`, `250|30`) the average is wrong: it prints 141 instead of 141.67, because the code uses floor division. With the input `sedan` (no colon) the program crashes with a ValueError.

Step 3: Completing test_ride_share_fare.py with Hypothesis

Each function has two fuzz tests, eight in total. A **no-crash test** feeds random Unicode text and `None`. A **property test** builds input that looks valid (cab:km, a CSV of fares, fare|promo) from fuzzed numbers including negatives, huge values, `nan` and `inf`, and checks that any result returned is finite and non-negative. Two extra checks test that the average is mathematically correct and that a promo never raises the fare.

Step 4: Running the tests on the buggy code

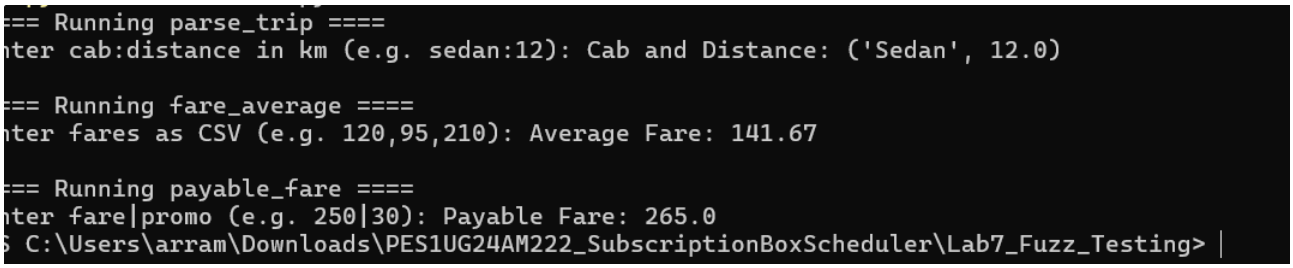
```
py -m pytest test_ride_share_fare.py
```

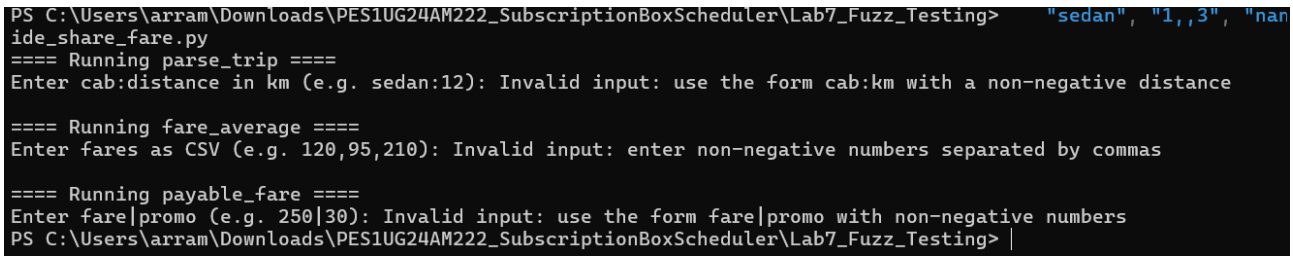


All 8 tests fail. Hypothesis shrank each failure to a minimal input, for example `fares=[0, 1]` giving an average of 0, and a promo of -1.0 raising the fare from 0.0 to 1.0.

Step 5: Fixing and running the application

```
py ride_share_fare.py
```

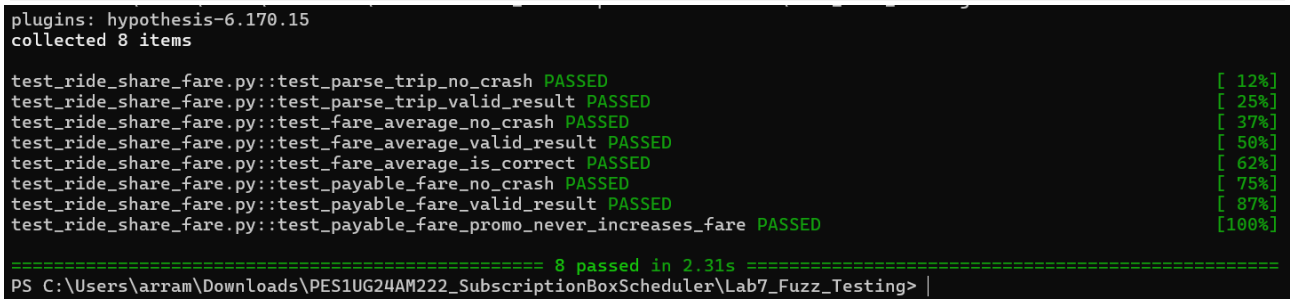




Valid input now gives the correct average, 141.67. Invalid input (`sedan`, `1,,3`, `nan|0`) prints an "Invalid input" message instead of crashing or returning nan.

Step 6: Re-running the tests on the fixed code

```
py -m pytest test_ride_share_fare.py -v
```



All 8 tests pass.

Summary: bugs found and fixed

Function	Bug found by fuzzing	Fix
parse_trip	Crashes on None, a missing ":" or extra ":", and an empty or non-numeric distance	Check the type and that there is exactly one ":"; convert numbers safely
parse_trip	Accepts negative, nan and inf distances, and an empty cab name	Reject non-finite or negative values and blank names
fare_average	Crashes on None, an empty string and empty items such as "1,,3"	Validate every item; return None if any is invalid
fare_average	Floor division gives wrong averages ([0, 1] gives 0)	True division, rounded to 2 decimals
payable_fare	Crashes on None, a missing "|" and non-numeric parts	Same input checks as above
payable_fare	"nan|0" returns nan; huge fares return inf	Reject non-finite input and results
payable_fare	A negative promo increases the fare ("100|-50" gives 168)	Reject negative fares and promos

Invalid input now returns `None`, and `main()` prints an "Invalid input" message instead of crashing.